

ED Triage & Patient Flow

Synthetic Data Case Study

Synthetic Australian Hospital Operations Dataset Bundle

Executive Summary

This case study walks through a sample analysis of the ED Triage & Patient Flow dataset — 45,000 synthetic emergency department presentations across 6 fictional facilities, using the Australasian Triage Scale (ATS 1–5). It demonstrates the kind of crowding, flow-time, and disposition analysis this dataset supports. All figures below are calculated directly from the dataset.

△ This is a synthetic dataset. Every figure below describes a simulated pattern, not a real-world clinical or operational finding.

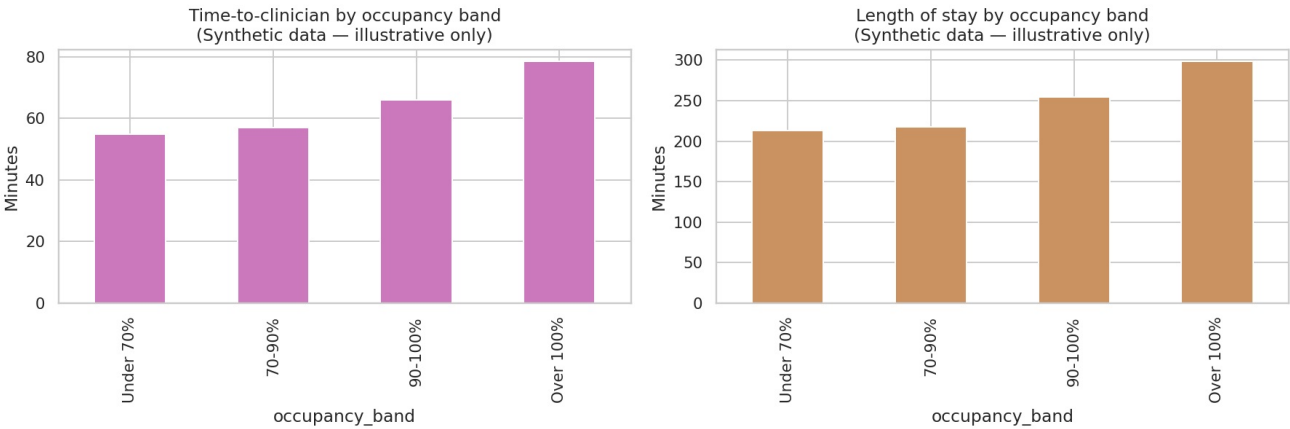
Dataset Snapshot

- 45,000 ED presentations, 16 columns
- 6 fictional facilities
- Bed-block rate: 2.52% of presentations
- Date range: 2025-01-01 to 2025-12-30

Key Findings (in this synthetic model)

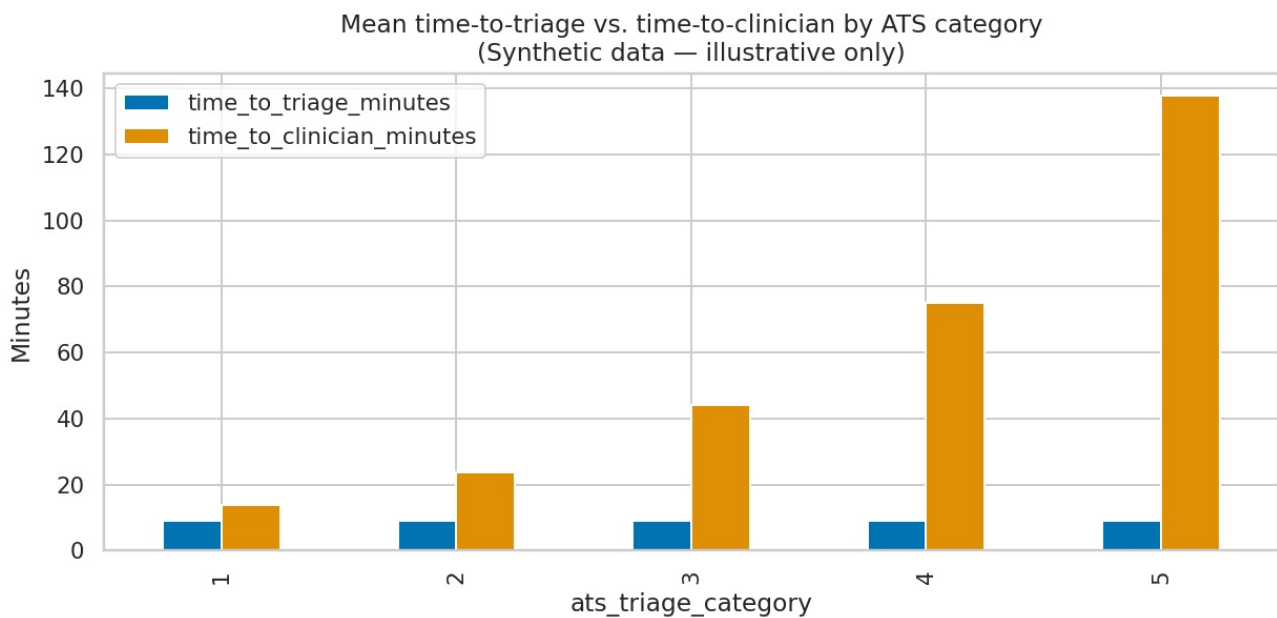
- 80.6% of presentations exceed their ATS-specific target wait time for time-to-clinician (targets: ATS1=0, ATS2=10, ATS3=30, ATS4=60, ATS5=120 minutes) — a modelled crowding effect built into the generator.
- Length of stay generally decreases with less urgency, with one designed exception: ATS1 (292.4 min) and ATS2 (273.2 min) show the longest modelled stays, reflecting resuscitation/stabilisation workup time, while ATS5 (192.3 min) sits slightly above ATS4 (175.1 min) — the least-urgent patients wait longest to be initially seen, even though their treatment once seen is quick.
- Arrival mode is linked to urgency by design: ambulance and police arrivals skew toward more urgent ATS categories, while walk-in arrivals skew toward less urgent categories.
- Bed-block is a rare but present modelled signal, tied to admission disposition and high department occupancy (over 85%).

Visual Highlights



Occupancy and crowding

Time-to-clinician and length of stay both rise with modelled department occupancy — occupancy can exceed 100% by design, representing an ED operating beyond funded capacity.



Time to triage and clinician by ATS category

Mean time-to-triage and time-to-clinician scale with ATS-specific target wait times.

Methodology Note

Presenting complaint and arrival mode both influence the ATS category distribution; time-to-clinician and length of stay scale with modelled department crowding and ATS-specific targets. These are designed relationships built into the data generator — not measurements of any real emergency department. Full methodology, v1.1 data-quality corrections, and limitations: see [methodology_bias_limitations.md](#).

Why This Dataset Is Useful

This dataset gives analysts, students, and QA/software testers a realistic-feeling, richly-structured ED operations dataset to practise on without any real patient privacy risk. It supports:

- SQL and Python flow-time and crowding-analysis exercises
- Power BI/Tableau dashboard-building practice (ATS mix, disposition, facility comparison)
- An educational segmentation/clustering exercise on operational variables
- Portfolio-ready case studies and dashboards — like this one

Restriction / Disclaimer

100% synthetic data. No real patients, clinicians, staff, or facilities are represented. Not clinically validated. For research, education, software testing, and AI/ML training/evaluation only. Full terms: [LICENSE_AND_ACCEPTABLE_USE.md](#).

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